
MGT 4050 Bikeshare Project

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Table of Contents



Background



Exploratory
Analysis



Models



Evaluation
Metrics



Findings



Recommendatio
ns

01

Background



About the Dataset

This bike-sharing dataset contains 17,414 records ranging from January 4, 2015 to January 3, 2017. It holds information surrounding the record of count of new bike shares with the factors temperature, weather conditions, humidity, wind speed, holiday status, weekend designation, and season.

Predictor Variables

- *"timestamp"*
- *"t1"*
- *"t2"*
- *"hum"*
- *"windspeed"*
- *"weathercode"*
- *"isholiday"*
- *"isweekend"*
- *"season"*

Response Variable

- *"cnt"*



Our Goals

0
1

To understand the relationship of the predictor variables on the response variable.

0
2

Create an appropriate prediction model to estimate number of new bike shares.

02

Exploratory Analysis



What feature is most associated with number of bike shares?

“weather_code10” (rain with thunderstorm) is most associated with number of bike shares.

- Estimated Coefficient: $-4.710e+02$
- P-Value: 0.048050 *

***Linear Regression Model ran with weather_code1 as the base case

```
> summary(lm_1)
```

Call:

```
lm(formula = cnt ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1926.3	-573.2	-244.6	294.9	6032.2

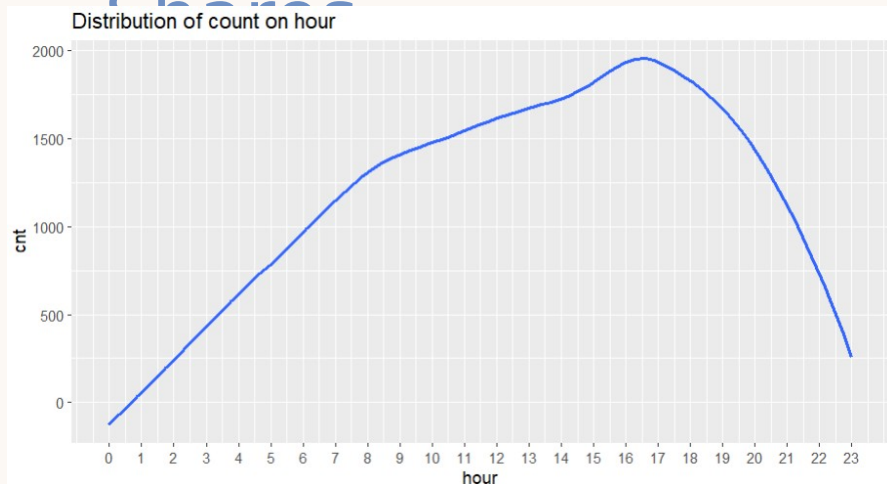
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.675e+03	7.095e+02	2.361	0.018258 *
timestamp	2.455e-07	4.968e-07	0.494	0.621156
t1	8.734e+01	1.009e+01	8.660	< 2e-16 ***
t2	-2.961e+01	8.269e+00	-3.581	0.000343 ***
hum	-2.441e+01	7.301e-01	-33.434	< 2e-16 ***
wind_speed	-7.926e+00	1.082e+00	-7.324	2.52e-13 ***
weather_code2	2.839e+02	1.971e+01	14.401	< 2e-16 ***
weather_code3	2.892e+02	2.172e+01	13.314	< 2e-16 ***
weather_code4	-1.554e+02	3.069e+01	-5.064	4.14e-07 ***
weather_code7	-1.523e+01	2.684e+01	-0.567	0.570508
weather_code10	-4.710e+02	2.382e+02	-1.977	0.048050 *
weather_code26	1.206e+02	1.265e+02	0.954	0.340336
is_holiday1	-2.200e+02	5.182e+01	-4.246	2.19e-05 ***
is_weekend1	-1.899e+02	1.617e+01	-11.741	< 2e-16 ***
season1	-5.629e+01	2.617e+01	-2.150	0.031537 *
season2	1.642e+02	2.985e+01	5.501	3.84e-08 ***
season3	2.283e+01	2.158e+01	1.058	0.290188
month	-2.187e+01	3.680e+00	-5.942	2.88e-09 ***
day	-3.551e+00	8.291e-01	-4.283	1.86e-05 ***
hour	2.889e+01	1.122e+00	25.757	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

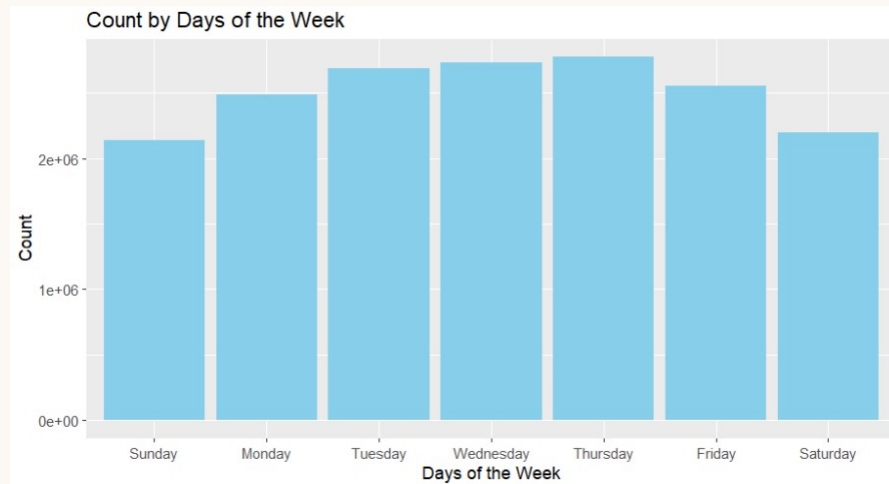
Residual standard error: 889.3 on 15114 degrees of freedom
Multiple R-squared: 0.3443, Adjusted R-squared: 0.3435
F-statistic: 417.7 on 19 and 15114 DF, p-value: < 2.2e-16

The Effect of Time on Number of Bike



The peak of number of bike shares usually occurs between 4-5PM.

→ Potential Reasons: end of workday, school dismissal, nicer weather, social engagements, reduced transit options

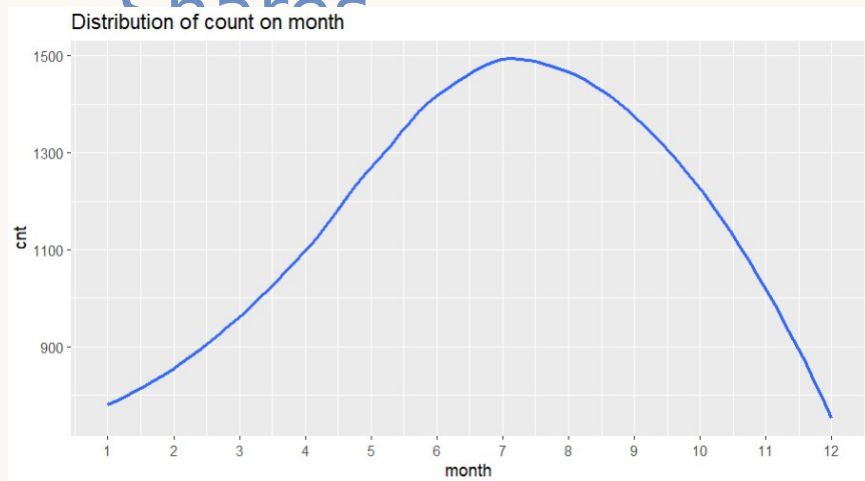


The number of bike shares is highest during the middle of the week.

→ Potential Reasons: midweek errands/pre-weekend preparation, crowded public transportation

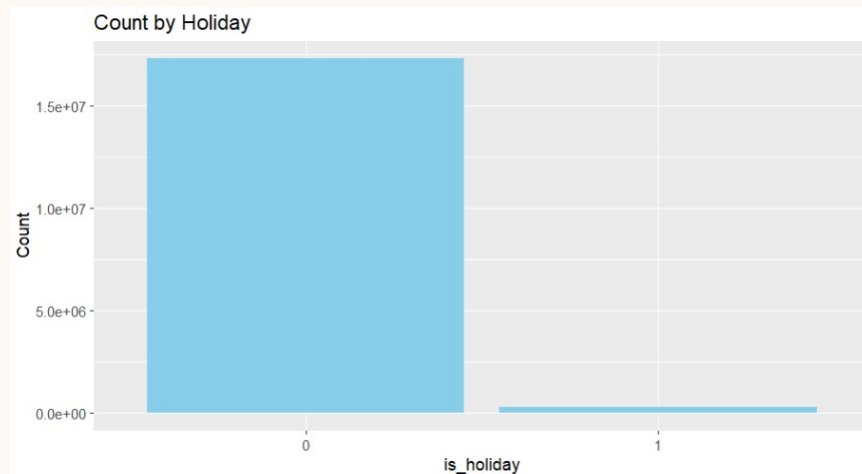
The Effect of Time on Number of Bike

Shares



The number of bike shares is highest in July.

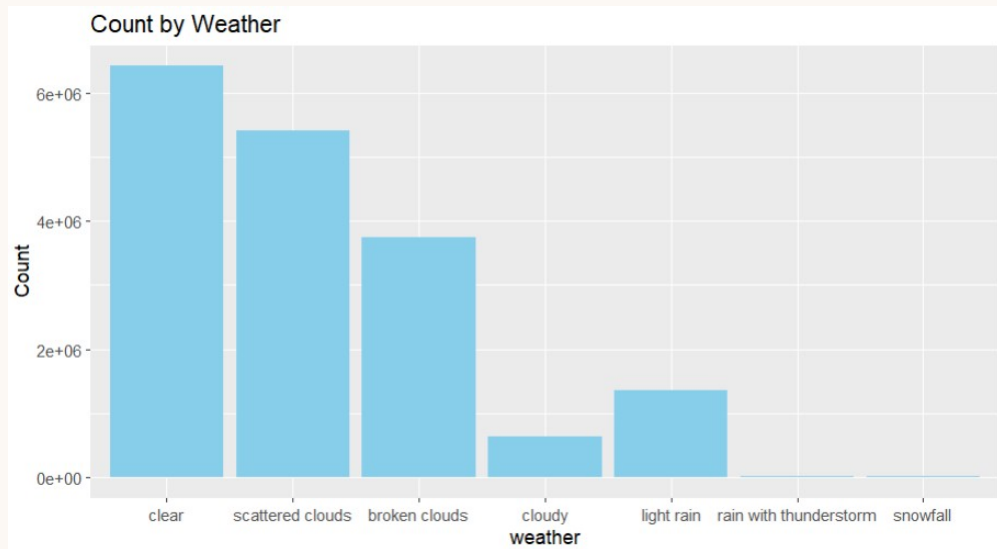
→ Potential Reasons: nicer weather, summer activities, tourism, health & fitness focus



People are much less likely to require bike sharing on holidays.

→ Potential Reasons: increased car/plane travel, family & social gatherings, reduced work & school commute

What is the relation between weather situations and the number of bike shares?



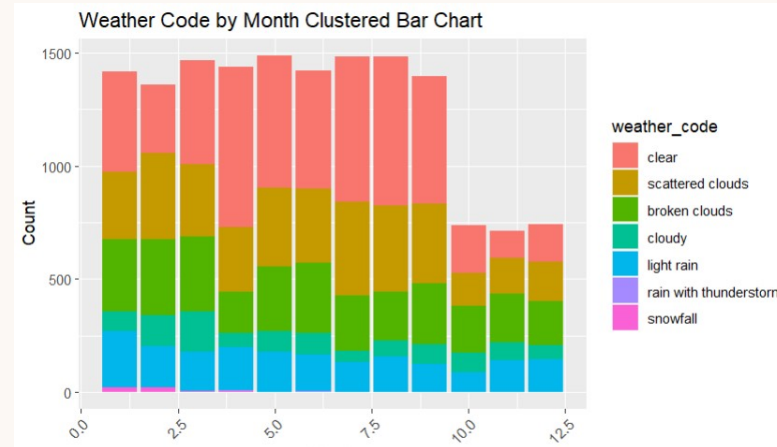
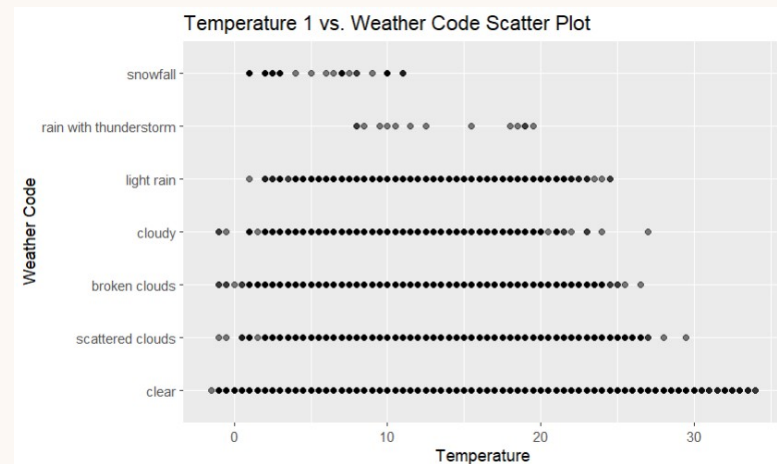
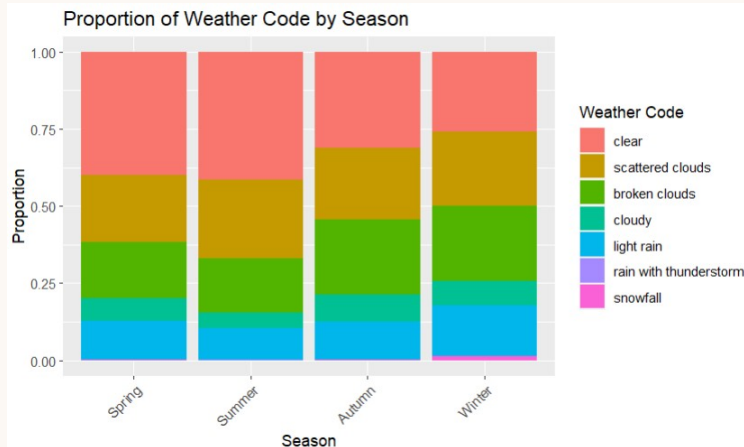
Number of bike shares is higher for weather situations with potential for sunshine.

→ More favorable bike riding conditions of sunlight, visibility, and warmer temperatures.

Bike sharing is significantly lower when there is rain with thunderstorms or snowfall.

→ Bike riding conditions are dangerous and uncomfortable during these weather types.

Temperature, season, and month do not appear to have a significant effect on weather_code. Therefore, we can assume that the variables are not highly correlated.



Key Takeaways

01

“weather_code10”, *“weather_code2”*, and *“weather_code3”* have the largest magnitude of impact on number of bike shares.

02

On average, the number of bike shares is highest in the afternoons and weekdays.

03

Clear to semi-clear weather prompts highest number of bike shares.

03

Models



Models Created

Linear Regression

Build Linear Regression model with all predictor variables and used backward selection to refine predictions of bike share count

Decision Trees

Utilize Decision Trees to predict bikeshare count while using Bagging and Random Forest approaches to improve accuracy of predictions

Does not predict exact "cnt" values!



Nonlinear Regression

Capture potentially nonlinear relationship between bike share count and other predictor variables to improve prediction accuracy beyond a linear regression model

Clustering & Logistic Regression

Group bike share data into clusters and utilize logistic regression to understand how the clusters relate to the count of bike shares

Linear Regression

Linear Regression:

- R-Squared = 34.42%
- Test MSE: 919,153.8
- Relationship does not seem to be linear, as demonstrated by Q-Q Plot and Residual Plot

```
Call:
lm(formula = cnt ~ . - timestamp - season3 - weather_code7 -
    weather_code26, data = final_data)
```

Residuals:

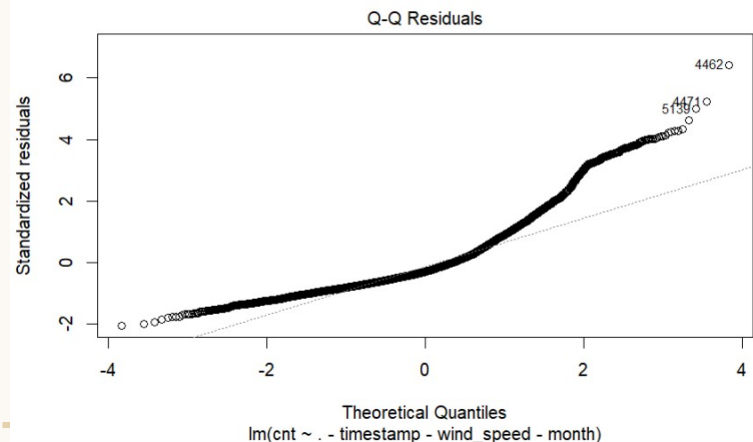
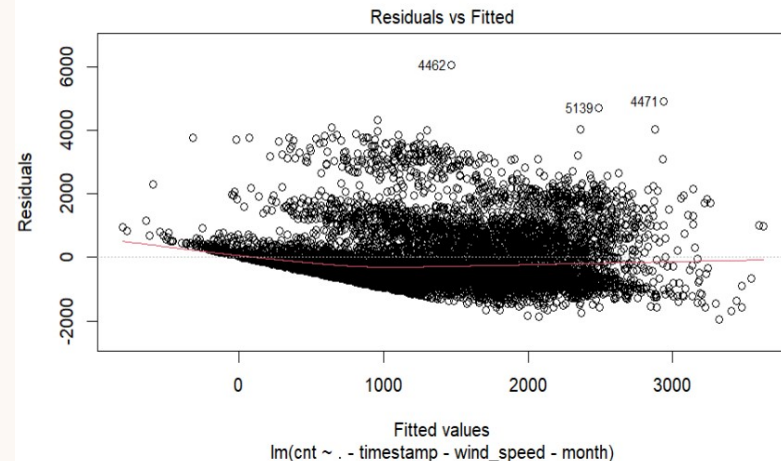
	Min	1Q	Median	3Q	Max
	-1927.3	-573.9	-244.8	297.9	6026.6

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2036.9918	72.9860	27.909	< 2e-16	***
t1	88.3248	9.9979	8.834	< 2e-16	***
t2	-30.8974	8.1994	-3.768	0.000165	***
hum	-24.4255	0.6493	-37.620	< 2e-16	***
wind_speed	-7.9740	1.0372	-7.688	1.58e-14	***
is_holiday1	-223.4493	51.7222	-4.320	1.57e-05	***
is_weekend1	-189.7988	16.1671	-11.740	< 2e-16	***
month	-21.5470	3.6563	-5.893	3.87e-09	***
day	-3.5413	0.8275	-4.280	1.88e-05	***
hour	28.9159	1.1141	25.954	< 2e-16	***
season1	-60.5690	25.5795	-2.368	0.017903	*
season2	156.7404	28.7232	5.457	4.92e-08	***
weather_code2	288.1271	18.4621	15.606	< 2e-16	***
weather_code3	294.1043	19.5511	15.043	< 2e-16	***
weather_code4	-152.1351	29.3870	-5.177	2.28e-07	***
weather_code10	-469.7749	237.9845	-1.974	0.048403	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 889.2 on 15118 degrees of freedom
Multiple R-squared: 0.3442, Adjusted R-squared: 0.3435
F-statistic: 528.9 on 15 and 15118 DF, p-value: < 2.2e-16



Polynomial Regression

Polynomial Regression to the 2nd Degree:

- R-squared Value: 34.84%

Best R-squared

Polynomial Regression to the 3rd Degree:

- R-squared Value: 35.08%
- Test MSE: 822,491.2

Polynomial Regression to the 4th Degree:

- R-squared Value: 33.09%

```
>
> predictions <- predict(test_model2)
>
> mse <- mean((data$cnt - predictions)^2)
> cat("MSE:", mse, "\n")
MSE: 822491.2
> |
```

Call:

```
lm(formula = cnt ~ t1_poly + t2 + hum_poly + wind_speed_poly +
    weather_code + season + hour, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2495.3	-546.8	-245.3	235.4	6110.9

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3177.71407	154.20214	20.607	< 2e-16 ***
t1_poly1	-57.95326	14.60119	-3.969	7.25e-05 ***
t1_poly2	2.77141	0.21830	12.695	< 2e-16 ***
t2	23.52701	9.12301	2.579	0.00992 **
hum_poly1	-55.30600	4.40846	-12.545	< 2e-16 ***
hum_poly2	0.23372	0.03232	7.232	5.00e-13 ***
wind_speed_poly1	23.12347	3.52481	6.560	5.55e-11 ***
wind_speed_poly2	-0.67350	0.08448	-7.972	1.67e-15 ***
weather_code2	317.15388	19.75856	16.051	< 2e-16 ***
weather_code3	306.40103	21.67208	14.138	< 2e-16 ***
weather_code4	-162.69069	30.53360	-5.328	1.01e-07 ***
weather_code7	-15.44560	27.18119	-0.568	0.56988
weather_code10	-412.07307	237.50462	-1.735	0.08276 .
weather_code26	-19.10872	126.43294	-0.151	0.87987
season1	-83.92225	25.71416	-3.264	0.00110 **
season2	95.86306	24.03689	3.988	6.69e-05 ***
season3	-37.45426	22.12021	-1.693	0.09044 .
hour	29.17961	1.11871	26.083	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 886.4 on 15116 degrees of freedom
(2 observations deleted due to missingness)
Multiple R-squared: 0.3484, Adjusted R-squared: 0.3477
F-statistic: 475.4 on 17 and 15116 DF, p-value: < 2.2e-16

Decision Trees

Original Decision Tree:

- Predictors: "hour", "t1", "is_weekend"
- Test MSE: 238,475.4

Pruned
Decision Tree
(18 leaf
nodes)

Fully Grown Tree (largetree):

- Predictors: all
- Test MSE: 132,841.6

Pruned Tree:

- Predictors: "hour", "t1", "is_weekend"
- Test MSE: 1,727,608

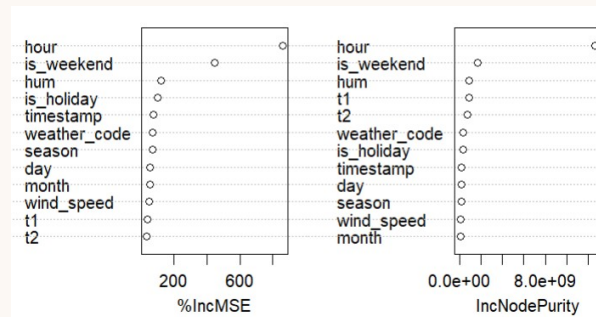
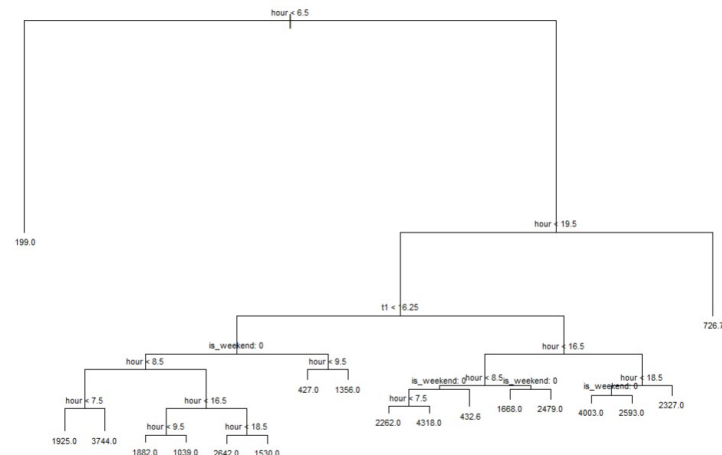
Bagging Method:

- Most Important Predictors: "hour", "is_weekend"
- Test MSE: 82,640.95

Best Test MSE

Random Forest Method:

- Most Important Predictors: "hour", "is_weekend"
- Test MSE: 124,954.5



Variable
Importance
using Bagging

04

Evaluation Metrics



Comparison of Test Set MSE

- Measured square error (MSE) is the most appropriate and commonly-used measure for regression analysis
 - We observed more complex models to underfit the data and yield a lower test MSE
 - (1) Large, fully grown tree (11 predictor variables) < Default tree (3 variables)
 - (2) Large, fully grown tree (77 terminal nodes) < Pruned tree (18 terminal nodes)
 - (3) Bagging (mtry = 11) < Random Forest (mtry = 5)
 - Causes:
 - The training data set is too large (~15,137 observances)
 - The training data set is not normalized
 - Insufficient predictor variables
-

“mtry = ”

- Indicates the number of predictor variables considered for each split of the tree
- $mtry \leq$ number of predictor variables: [1, 11]

```
> bagMSE  
[1] 82640.95
```

```
> rfMSE  
[1] 121539.8
```

```
> rf <- randomForest(cnt ~ ., data = data, subset = train, mtry=5, importance = TRUE)
```

```
> bag <- randomForest(cnt~., data= data, subset = train, mtry=11, importance = TRUE)
```

- ★ The larger the mtry value, the smaller the testMSE
- Usually, when $mtry < \#$ of predictor variables, the testMSE is smaller
 - Random Forest
 - Reduces overfitting
- Although $mtry = \#$ of predictor variables, our tree is underfit due to the size of the dataset
 - Training a subset of predictor variables causes reduced predictive performance and loss of information

“ntree = ”

- Indicates the number of trees to include in the ensemble methods
- No upper limitation
- Default is 500 trees
- Beyond a certain number of trees, the benefit of prediction accuracy becomes marginal

```
> bag <- randomForest(cnt~., data= data, subset = train, mtry=11,importance = TRUE)
```

```
> bag1 <- randomForest(cnt~., data= data, subset = train, mtry=11, ntree = 400, importance = TRUE)
```

```
> bag2 <- randomForest(cnt~., data= data, subset = train, mtry=11, ntree = 1000, importance = TRUE)
```

- ★ The larger the ntree value, the smaller the testMSE
- Pro: Improve predictive performance
- Con: Dramatic increase in processing time and computational power requirements

```
> bagMSE  
[1] 82640.95  
ntree = 500
```

```
> bagMSE1  
[1] 82652.47  
ntree = 400
```

```
> bagMSE2  
[1] 82372.69  
ntree = 1000
```

05

Findings



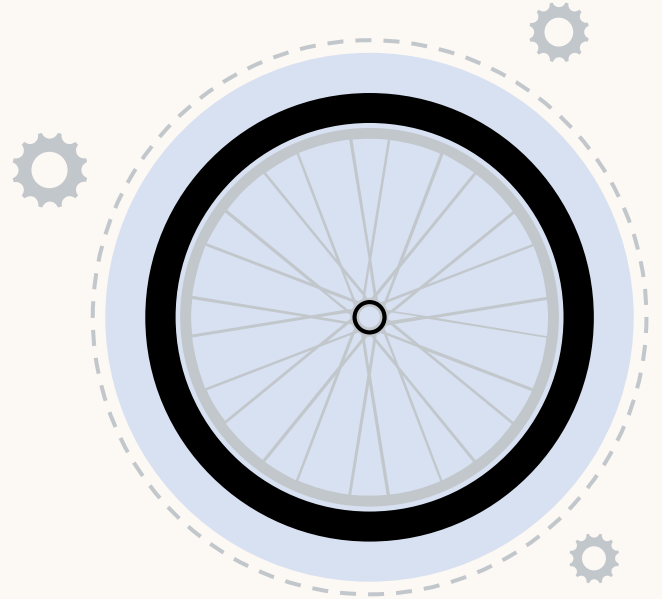
Critical Insights from Analysis

Data size:

- Due to the fact that the large data size, it is a big challenge for the computational power
- We need better robust data handling and analysis techniques for accurate prediction

Important variables:

“Hour” and “is_weekend”



06

Recommendati
on



Tracking and Understanding Usage

- **GPS tracking:**
Track routes, locations and usage time.

Gathering Data and Engaging Users

- **User surveys:** collect users feedback on the purpose of using share bi
- **Targeted promotions:** run promotions during peak hours

Optimizing Bike Share Usage

- **Efficient Rebalancing & Focus on peak hours**
- Implement a dynamic scheduling system for rebalancing
- Optimize resource allocation based on peak hours.
Ensure that bike availability matches demand patterns.



Thank You!

